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Problem 1 (Transformation of random Variables)

a) $Z \sim U(-a,a)$; $X \sim N(0,\sigma^2)$

We have $P_r(\alpha < Z < \beta) = P_r(T(\alpha) < X < T(\beta))$

$$\Rightarrow \int_\alpha^\beta \frac{1}{2a}\, dz = \int_{T(\alpha)}^{T(\beta)} \frac{1}{\sigma\sqrt{2\pi}} e^{\left(-\frac{x^2}{2\sigma^2}\right)} dx \Rightarrow dz = 2a \cdot \frac{1}{\sigma\sqrt{2\pi}} e^{\left(-\frac{x^2}{2\sigma^2}\right)} dx$$

$$\Rightarrow \frac{dz}{dx} = 2a \underbrace{\frac{1}{\sigma\sqrt{2\pi}} e^{\left(-\frac{x^2}{2\sigma^2}\right)}}_{f_1(x)} = \frac{d}{dx} 2a F_X(x) \Rightarrow \underset{[-a,a]}{z} = \underbrace{2a F_X(x)}_{[0,1]} + c$$

$[0,2a] \Rightarrow c = -a$

So we find $z = 2a F_X(x) - a$

$$\Leftrightarrow F_X^{-1}\left(\frac{z+a}{2a}\right) = x \Rightarrow X = F_X^{-1}\left(\frac{Z+a}{2a}\right) = T(Z)$$

$[-a,a] \to \mathbb{R}$

$T: Z \mapsto F_X^{-1}\left(\frac{z+a}{2a}\right)$

$\sim U(-a,a)$ $\sim N(0,\sigma^2)$

- $\check{Z} \sim U(0,1)$; $X \sim \mathcal{L}(0,b)$

$$F_X(x) = \int_{-\infty}^{x} \frac{1}{2b} e^{-\frac{|x-0|}{b}} dx = \begin{cases} \frac{1}{2}\int \frac{1}{b} e^{\frac{x}{b}} dx = \frac{1}{2} e^{\frac{x}{b}} & x \le m \\ \frac{1}{2}\int \frac{1}{b} e^{-\frac{x}{b}} dx = -\frac{1}{2} e^{-\frac{x}{b}} & x \ge m \end{cases}$$

$$\int 1\, d\check{z} = \check{z} = \pm e^{\frac{\pm x}{b}} \Rightarrow -b \cdot \log(2\check{z}) \cdot \operatorname{sign}(\check{z} - \tfrac{1}{2}) = X$$

$$\Rightarrow -b \cdot \operatorname{sign}(\check{Z} - \tfrac{1}{2}) \cdot \log\left(1 - 2 \cdot \left|\check{Z} + \tfrac{1}{2}\right|\right) = X$$

$$\check{Z} = \frac{Z+a}{2a} \Rightarrow \underset{\sim \mathcal{L}(0,b)}{X} = -b \operatorname{sign}\left(\frac{Z+a}{2a} - \frac{1}{2}\right) \cdot \log\left(1 - 2\left|\frac{Z+a}{2a} - \frac{1}{2}\right|\right)$$

b)

- We have $Y \sim U(-a,0)$ $Z \sim N(m,\sigma^2)$ $X \sim \mathcal{L}(m,b)$

$$T_1: \underset{\sim U(-a,0)}{y} \mapsto \underset{N(m,\sigma^2)}{F_Z^{-1}\left(\frac{y+a}{2a}\right) + m} \Rightarrow Y = 2a F_Z(Z-m) - a \sim U(-a,0)$$

$$T_2: y \mapsto -b \operatorname{sign}\left(\frac{y+a}{2a} - \frac{1}{2}\right) \cdot \log\left(1 - 2\left|\frac{y+a}{2a} - \frac{1}{2}\right|\right) + m \sim \mathcal{L}(m,b)$$

$$\Rightarrow -b \operatorname{sign}\left(F_Z^{-1}(z-m) - \tfrac{1}{2}\right) \cdot \log\left(1 - 2\left|F_Z^{-1}(z-m) - \tfrac{1}{2}\right|\right) \sim \mathcal{L}(m,b)$$

$$T: Z \mapsto -b \operatorname{sign}\left(F_Z^{-1}(z-m) - \tfrac{1}{2}\right) \cdot \log\left(1 - 2\left|F_Z(z-m) - \tfrac{1}{2}\right|\right)$$